Classification of eating behaviors in unconstrained environments

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Abstract. Obesity and its numerous devastating consequences are on the rise globally. While widespread tactics to fight against obesity often focus on healthy eating, how the food is consumed is oftentimes overlooked even though convincing evidence attests that merely eating slowly and properly chewing one's meal significantly reduce obesity. This research introduces a method that recognizes common human actions during mealtime viz., food chewing, food swallowing, drink swallowing, and talking. The proposed system is unobtrusive. It uses a cheap and small bone conduction microphone to collect intra-body sound and a smartphone that processes sound and provides feedback in real-time so that it can be used conveniently in daily life. Our proposed approach achieves similar performances ($specificity = 98.0\%, precision = 83.8\%, recall = 91.7\%$, $Jaccard\,score = 0.89, F_1\,score = 87.2\%$ and $MCC = 0.85$) as those achieved by the most recent state of the art models even though our system uses modest machine learning models.

Keywords: eating quantification· chewing· swallowing· sound analysis, activity recognition · free-living conditions

1 Introduction

Healthy eating is often hailed as the most reliable method to fight against obesity. However, while a healthy diet certainly plays an influential role in weight loss, how the food is consumed is an often overlooked obesity-fighting mechanism. Indeed, accumulating evidence shows that merely eating slowly and properly chewing one's meal reduce obesity significantly. For instance, an up-to-date systematic review [1] on the topic concluded that inadequate mastication highly correlates with obesity. Likewise, a large scale study (N=92,363 individuals) conducted between 2005 and 2013 found that those with slow eating habits have, among other numerous benefits, a lower BMI (22.3±4.0), and a lower obesity rate (21.5%).

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In contrast, those with fast eating habits have a much higher BMI (25.0 ± 4.4) and a higher obesity rate (44.8%) [2].

There exist diverse food intake monitoring mechanisms. The simplest strategies consist of manual self-reports. However, self-report is imprecise because it is biased and subjective [3, 4]. The germane state of the art [5–7] proposes wearable devices as a more accurate alternative. Unlike manual mechanisms, wearable gadgets automatically monitor the dietary habits of the wearers; thus, they are immune to subjective bias. Moreover, they are more convenient and can provide real-time feedback to their users [6]. Although automatic food monitoring is still an open problem to solve [5, 8, 9], many solutions have been introduced and are extensively reviewed in e.g., [5, 6]. For example, [8] proposed an earpiece that automatically monitors eating behavior in free-living conditions. Their system is based on an off-the-shelf contact microphone and has an $F_1 score$ that exceeds 77.5% for eating detection. Recently, [10] presented a framework for in-the-wild modeling of eating behavior using inertial measurement unit (IMU) signals that are recorded from smartphones. The proposed approach uses both convolutional and recurrent layers that are trained simultaneously. This model outperforms the state-of-the-art in detecting food bites ($F_1 score = 92.3\%$).

In our previous studies [11–14], we proposed a method that uses a cheap and small bone conduction microphone to automatically monitor eating behaviors. Our research was, however, restricted in scope. This paper extends the following improvements to our earlier research and to other existing relevant studies:

1) All experiments were conducted in free-living environments (e.g., at home when the subjects were eating with their families) to capture the complexity of the real-world.
2) The proposed approach recognizes five different eating activities (i.e., chewing, food swallowing, drink swallowing, talking and other noises).
3) There was no constraint on the type of food to eat and the participants were urged to have whatever they usually eat or drink.
4) The proposed approach achieves comparable performance ($accuracy = 97.5\%$, $precision = 83.8\%$, $recall = 91.7\%$, $Jaccard\ score = 0.89$, $Kappa\ score = 0.88$ and $F_1\ score = 87.2\%$) to those achieved by the most recent state of the art model proposed in [10] even though our proposed method uses much simpler machine learning models.

2 METHODS

2.1 Data collection

Sixteen subjects joined the experiments. All experiments were conducted in free-living environments to capture the complexity of the real-world. For instance, some dietary sounds were recorded at home when the subjects were having supper with their families. Others were collected in a university cafeteria during lunch with colleagues. Additionally, there was no constraint on the type of food and the subjects were urged to have whatever they usually eat.

The dietary sounds were recorded ($Fs = 8\,\text{kHz}$) using a commercial bone conduction microphone (Motorola Finiti HZ800 Bluetooth Headset, Motorola co. Ltd.) that is attached to a subject's ear. The microphone sends the recorded sounds to a smartphone via Bluetooth. The sounds are then saved for further analysis. At the same time, another smartphone records the subject's mouth and throat motion as shown in Fig. 1.

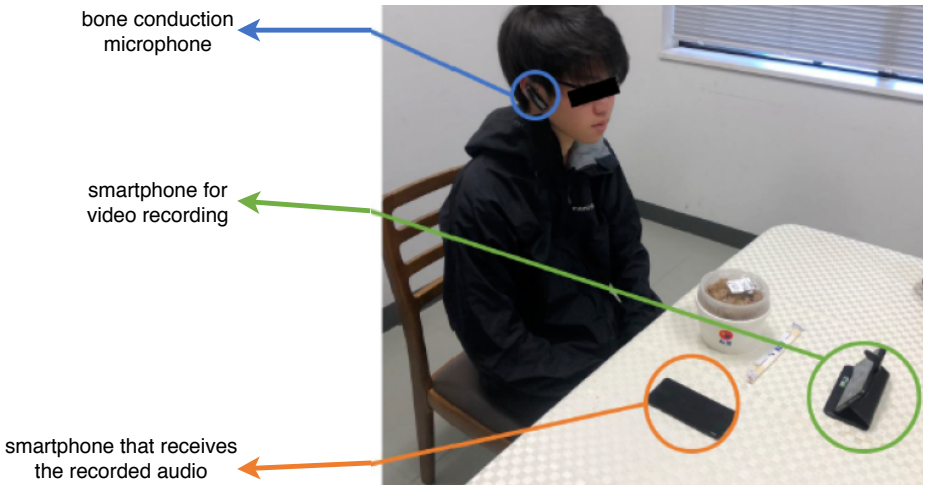


FIG. 1: Experimental setup of a typical dietary sound collection
The experiments were conducted in a free-living environment when the subjects dinned with e.g., their family. The dietary sounds were recorded using a bone conduction microphone. One smartphone was used to receive, via Bluetooth, the recorded dietary sounds. Another is used to record a video of the subject for data annotation purposes.

2.2 Audio data labeling

The collected dietary sounds are segmented and then labeled using Praat[1], a free software for speech analysis in phonetics that, among others things, allows speech labeling and segmentation. Labels were manually added by looking at the video which were recorded when the subjects were having their meals. The following five labels were added:

- *Chewing* —this label is assigned to any sound segment corresponding to when the subjects were crushing solid food with their teeth.
- *Food swallowing* —when the subjects swallow the chewed solid food but not when swallowing liquids.
- *Drink swallowing* —when the subjects swallow liquids but not solid foods.
- *Talking* —when the subjects are speaking.
- *Other sounds* —this label is assigned to sounds segments when other sounds other than the aforementioned are recorded.

[1] http://www.fon.hum.uva.nl/praat/

2.3 Feature computation

A total of 26 features (Table I) were extracted from the recorded dietary sounds. The *chroma features* capture harmonic and melodic characteristics of music and they are robust to changes in timbre and instrumentation. They are also used for audio-matching. A chroma vector is a 12-element feature vector indicating how much energy is contained in each of the 12 pitch classes (C, C#, D, D#, E, F, F#, G, G#, A, A#, B). The 12-dimension chroma vector $v_c(t)$ is computed from a logarithmic Short-Time Fourier Transform (STFT) power spectrum $\Psi_p(f,t)$ [15, 16] (Eq. (1)) and constitutes a compact representation of a sound [17].

TAB. I: Summary of the extracted features

Description	Number of features
Mean of chroma vector	1
Root mean square energy	1
Spectral centroid	1
Spectral bandwidth	1
Spectral roll off	1
Zero crossing rate	1
Mel-frequency cepstral coefficients	20

$$v_c(t) = \sum_{h=Oct_L}^{Oct_H} \int_{-\infty}^{\infty} BPF_{c,h}(f)\Psi_p(f,t)df \tag{1}$$

$BPF_{c,h}(f)$ represents a band-pass filter (Eq. (3)) which passes the log-scale frequency $F_{c,h}$ (Eq. (2)) of pitch class c and is applied from low (Oct_L) to high (Oct_H) octaves.

$$F_{c,h} = 1200h + 100(c-1) \tag{2}$$

$$BPF_{c,h} = \frac{1}{2}\left\{1 - \frac{cos\left[2\pi\left(f - \left(F_{c,h} - 100\right)\right)\right]}{200}\right\} \tag{3}$$

The energy $E_s(t)$ of a continuous signal $x(t)$ and a discrete signal $x(n)$ corresponds to area under their squared magnitude (Eq. (4) and Eq. (5)). For audio signals, that roughly corresponds to the loudness of the signal.

$$E_s(t) = \int_{-\infty}^{\infty} |x(t)|^2 dt \tag{4}$$

$$E_s(t) = \sum_{n=-\infty}^{\infty} |x(n)^2| \tag{5}$$

The *Root Mean Square Energy (RMSE)* of an N samples discrete signal $s(n)$ is defined as the square root of the average of the sum of all samples n (Eq. (6)).

$$RMSE = \sqrt{\frac{1}{N}\sum_{N}|s(n)|^2} \tag{6}$$

The *spectral centroid (SC)* indicates which frequencies the energy of the spectrum is centered upon, i.e., it expresses where the center of gravity of the spectral energy resides [17] (Eq. (7)).

$$f_c = \frac{\sum_k S(k)f(k)}{\sum kS(k)} \tag{7}$$

where $S(k)$ is the spectral magnitude and $f(k)$ is the frequency at bin k.

The *Spectral bandwidth* (Eq. (8)) defines the order-p spectral bandwidth and is used to distinguish tonal sound from noise-like sounds [17].

$$f_b = \left(\sum_k S(k)\left(f(k) - f_c\right)^p\right)^{1/p} \tag{8}$$

where $S(k)$ is the spectral magnitude at frequency bin k, f(k) is the frequency at bin k, and f_c is the spectral centroid. When $p = 2$, this is like a weighted standard deviation.

The *Spectral roll-off* (Eq. (9)) of a point i is the frequency below which a specified percentile of the total spectral energy lies. It measures the skewness of the shape of the spectral [17, 18] and can be used to differentiate voiced from unvoiced speech because the former has its energy concentrated in high-frequencies while the latter has more energy in its lower bands [18].

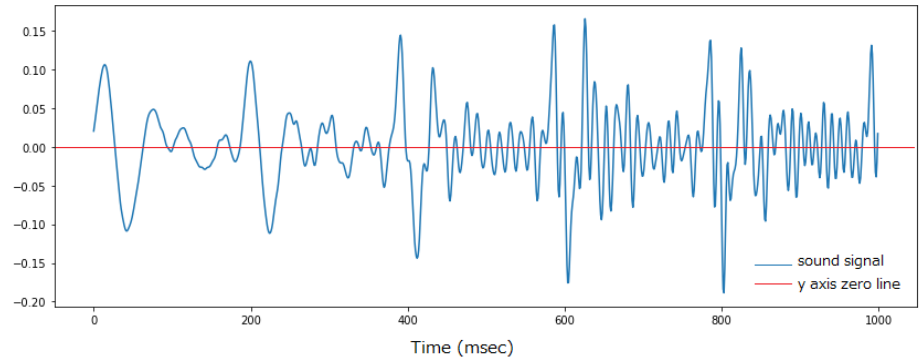
$$\sum_{k=b_1}^{i} s_k = K\sum_{k=b_1}^{b_2} s_k \tag{9}$$

where s_k is the spectral value at bin k, b_1 and b_2 are the band edges, in bins, over which to calculate the spectral spread. K is the percentage of total energy contained between b_1 and i.

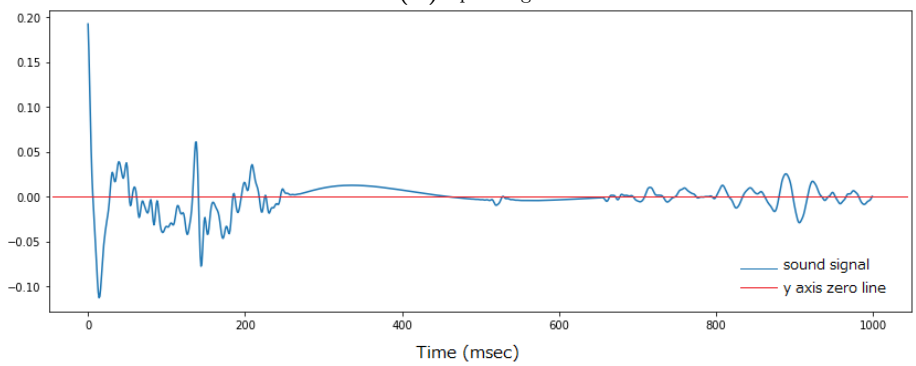
Zero crossing rate (ZCR) (Fig. 2) indicates the number of times that a signal crosses the horizontal axis and it provides a rough estimate of the dominant frequency in the sound [17].

Mel frequency cepstral coefficients (MFCC) are popular feature for sound classification. MFCC take into account humans' perception of sound and model the sound's spectral energy distribution accordingly [19]. Indeed, it is reported that humans' perception of a frequency f in a sound follows a non-linear frequency scale [19, 20]—the scale commonly known as the "mel" scale (Eq. (10))

$$f_{mel}(f) = 2595log_{10}\left(1 + \frac{f}{700}\right) \tag{10}$$



(a) speaking



(b) food swallowing

FIG. 2: A typical zero crossing rate of a talking and a swallowing sound

The MFFC are fundamental acoustic feature for speech and speaker recognition applications [19] and are calculated as follows. First, a sound is windowed and split into frames $\{y(n)_{n=1}^{N}\}$. Next, an M point Discrete Fourier Transform (DFT) is used to compute the power spectrum $Y(k)$ of each frame $y(n)$ (Eq. (11)).

$$\left|Y(k)\right|^2 = \left|\sum_{n=1}^{M} y(n).e^{\left(\frac{-j2\pi.n.k}{M}\right)}\right| \tag{11}$$

Finally, the power spectrum $Y(k)$ is converted into logarithmic domain and used to calculate the MFFC coefficients C_m (Eq. (12)).

$$C_m = \sqrt{\frac{2}{Q}} \sum_{l=0}^{Q-1} log\Big[e(l+1)\Big] cos\left[\frac{m\pi}{Q}\left(\frac{2l-1}{2}\right)\right] \tag{12}$$

where $0 \leqslant m \leqslant R-1$, R being the desired number of ancestral features; and in our case, $R = 20$.

2.4 Feature engineering

Two transformations was applied to the newly calculated dataset. The objective was to change the data distribution of of some features that were heavily skewed in order to potentially improve the performance of machine learning (ML) models. Box-Cox [21] and Yeo and Johnson [22] transformations were applied to the skewed features.

$$y(\lambda) = \begin{cases} \frac{y^{\lambda}-1}{\lambda}, & \text{when } \lambda \neq 0 \\ log(y), & \text{when } \lambda = 0 \end{cases} \tag{13}$$

The Box-Cox transform (Eq. (13)) improves the properties of the data and results in a symmetric dataset with a Gaussian distribution. The Yeo and Johnson (Eq. (14)) transformation leads to similar benefits and, unlike the Box-Cox transform, can be applied to negative values.

$$y(\lambda) = \begin{cases} \frac{(y+1)^{\lambda}-1}{\lambda}, & \text{when } \lambda \neq 0, \quad y \geqslant 0 \\ log(y+1), & \text{when } \lambda = 0, \quad y \geqslant 0 \\ \frac{(1-y)^{2-\lambda}-1}{\lambda-2}, & \text{when } \lambda \neq 2, \quad y < 0 \\ -log(1-y), & \text{when } \lambda = 2, \quad y < 0 \end{cases} \tag{14}$$

Additionally, new features (e.g., square, square root, cubit root, logarithms and inverse of each feature) were generate to improve the performance of the model.

2.5 Dataset oversampling

The resulting dataset is intrinsically unbalanced (Table II) because some dietary sounds are produced sporadically. For example, chewing generally takes a longer duration compared to drinking. Moreover, some people are fast eater while others eat slow and savor their meals.

TAB. II: Samples of each label in the dataset

label	number of samples
chewing	2001
talking	554
food swallowing	119
drink swallowing	82
other sounds	200

When applying ML algorithms on imbalanced datasets, the resulting models tend to downplay the importance of the minority classes. This problem is addressed by oversampling the minority classes using the borderline-synthetic minority over-sampling technique (Borderline-SMOTE) [23]. This method is an extension to the synthetic minority over-sampling technique (SMOTE) [24]. For each minority class X_j, SMOTE select one random seed sample $X_j^{'}$ from its k nearest neighbors and a new synthetic sample X_{synth} is interpolated.

While the SMOTE method works well for many datasets, it creates noise that decrease the performance of many ML models. Indeed, during training, in attempt to achieve the best performance, most ML algorithms learn the boundary of each class as exactly as possible. Consequently, the samples at the borderline are prone to misclassifications compared to those far away from the boundary because they are located where different classes may overlap [23]. Consequently, the sample at the decision boundary needs a special treatment to avoid the potential misclassifications. Unlike SMOTE —which oversamples all samples of the minority classes—the Borderline-SMOTE oversamples only the samples of the minority classes that are located on the boundary of the decision boundary.

Prior to applying the oversampling method, 25% of the samples in the dataset was randomly selected and reserved for testing the performance of the ML models. The remaining 75%, the training set, was oversampled using the Borderline-SMOTE and used to train various predictive ML models.

2.6 Feature selection

The feature engineering resulted in a total 958 features. There is a need to reduce this number to minimize overfitting and to avoid the curse of dimensionality. The following feature selection approaches were applied in a sequence. First, we removed features with unique values. These features would be practically useless for any ML model because they have a zero variance. Second, we removed the features that have a zero predictive importance, i.e., these features, have no information gain according to a gradient boosting machine (GBM) learning model. We also remove any features in the 5% lowest feature importance. Third, we remove highly correlated features. We computed the correlation between features and removed one feature of each pair whose correlation coefficient $c > 0.9$. Finally, a recursive feature elimination (RFE) [25] is used to rank and select the most optimum number of features. In the end, only 301 features were selected.

2.7 Model training and evaluation

A Tree-based pipeline optimization mechanism [26] was used to optimize (e.g., feature selection and hyperparameter tuning) the ML pipeline and to maximize the classification accuracy. Four auspicious pipelines based on an Extreme Gradient Boosting[1], a K-nearest Neighbors[2], a Random Forest[3] and Light Gradient Boosting Machine[4] models were evaluated. Two types of models were trained. The first model was trained on the original dataset before re-sampling and the second model was trained on the oversampled dataset. In all cases, the result reported in this study are based on the performance of the models when validated on the test set that was set aside before any oversampling.

[1] $XGBClassifier(colsample_bytree = 0.5, \gamma = 0.7, depth = 4, subsample = 0.4)$
[2] $KNeighborsClassifier(k_neighbors = 14, weights =$ "distance"$)$
[3] $RandomForestClassifier(depth = 24, max_features =$ "log2"$)$
[4] $LGBMClassifier(colsample_bytree = 0.7, depth = 32, num_leaves = 70, \alpha = 0.5)$

TAB. III: Performance of the models trained on the oversampled dataset

	XGBoost	KNeighbors	RandomForest	LGBM
Accuracy	97.29	96.32	97.56	97.62
Balanced accuracy	90.19	89.16	91.69	90.12
$F_\beta(\beta = 0.5)$	0.83	0.79	0.85	0.84
$F_\beta(\beta = 1)$	85.47	82.64	87.27	86.37
Jaccard-score	0.88	0.84	0.89	0.89
Geometric mean	94.69	92.78	95.18	95.04
Precision	81.95	78.10	83.83	83.41
Recall	90.19	89.16	91.69	90.12
Specificity	97.88	97.12	98.08	98.02
Matthews correlation coefficient	0.83	0.79	0.85	0.84

3 RESULTS AND DISCUSSION

Table III compares the performance of the models trained on the oversampled dataset and tested on the reserved unaltered (i.e, non-resampled) test samples. Although all the models performed quite well, the $RandomForest$ model slightly outperformed others. All the performance reported henceforth are based on the performance of the $RandomForest$ pipeline. On the contrary—and as expected—the models trained on the non-resampled dataset performed poorly (Table IV) compared to the models trained on the oversampled dataset (Table V). The models trained on the non-resampled dataset have adequate performance in predicting when the subjects were chewing or talking but have very poor performance in predicting when the subjects were swallowing drinks or when they were swallowing food (Table IV). On the contrary, although the model trained on the oversampled dataset sometimes confused food and drink swallowing for chewing (Fig. 4), in general, it consistently performed well ($specificity > 94.0\%, missrate < 17\%$ and $F_1 > 74.0\%$, and $MCC > 0.76$) in distinguishing all classes (Table V).

		Predicted values	
		FALSE	TRUE
Actual values	FALSE	True Negatives (TN)	False Positives (FP)
	TRUE	False Negatives (FN)	True Positives (TP)

FIG. 3: Confusion matrix

The performance of a model on an imbalanced dataset is normally evaluated in terms of its confusion matrix (Fig. 3). To test the performance of a classification

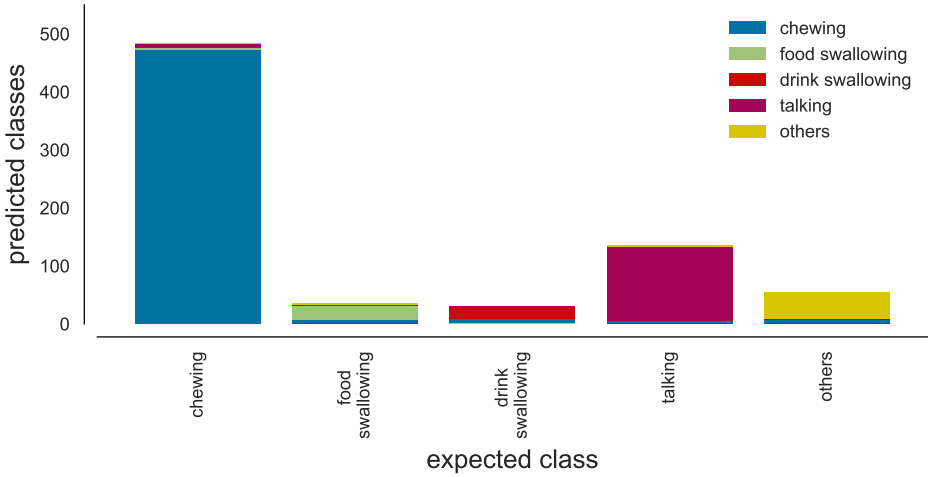


FIG. 4: Dietary sounds class labels prediction error
The model performs well in segregating the various dietary sounds labels. Nevertheless, it occasionally confuses food and drink swallowing for food chewing

model that is trained on an imbalanced dataset, the *accuracy* (Eq. (15)) is usually deceitful because it gives equal weight to false positives (FP) and false negatives (FN) [27]. On the contrary, a balanced accuracy (Eq. (16)) gives equal weight to FP and FN by computing the average of positive class instances that are correctly predicted and that of negative instances that are correctly classified.

$$accuracy = \frac{TP + TN}{TP + FP + TN + FN} \tag{15}$$

$$balanced\,accuracy = \frac{TP}{2(TP + FN)} + \frac{TN}{2(TN + FP)} \tag{16}$$

In Table III, one must note that, while the accuracy is high ($accuracy = 97.56\%$), the balanced accuracy is relatively lower ($accuracy = 91.69\%$), which implies that the model performs modestly lower than what an *accuracy* alone would have suggested.

The strong performance of the model trained on the oversampled dataset is further attested by its Receiver Operating Characteristic (ROC) and its Area Under the Curve (Fig. 5). The ROC evaluates the trade-offs between the model's sensitivity and its specificity. It computes the model's performance at various error rates and helps assess the percentage of samples that will be properly predicted for a given *FP/(TN+FP)*. Fig. 5a shows that the model trained on the non-resampled dataset performs poorly because many positive and negative instances are incorrectly classified. It is important to note that some classes are misclassified more than others. For instance, the model predicts reasonably well when the subjects were talking and chewing but the model achieves a much lower performance in predicting when the subjects were swallowing drinks. On the

TAB. IV: Performance of the model trained on the non-resampled dataset

	chewing	food swallowing	drink swallowing	talking	other sounds
Sensitivity	97.60	3.33	4.76	82.61	38.00
Specificity	64.02	99.58	100.00	98.17	97.68
Precision	85.02	25.00	100.00	91.20	54.29
Miss Rate	2.40	96.67	95.24	17.39	62.00
Accuracy	86.74	95.67	97.29	95.26	93.64
$F_\beta(\beta = 1)$	90.88	5.88	9.09	86.69	44.71
Matthews correlation coefficient	0.69	0.08	0.22	0.84	0.42

TAB. V: Performance of the model trained on the oversampled dataset

	chewing	food swallowing	drink swallowing	talking	other sounds
Sensitivity	94.40	86.67	90.48	94.93	92.00
Specificity	95.40	98.87	98.61	98.67	98.84
Precision	97.72	76.47	65.52	94.24	85.19
Miss Rate	5.60	13.33	9.52	5.07	8.00
Accuracy	94.72	98.38	98.38	97.97	98.38
$F_\beta(\beta = 1)$	96.03	81.25	76.00	94.58	88.46
Matthews correlation coefficient	0.88	0.81	0.76	0.93	0.88

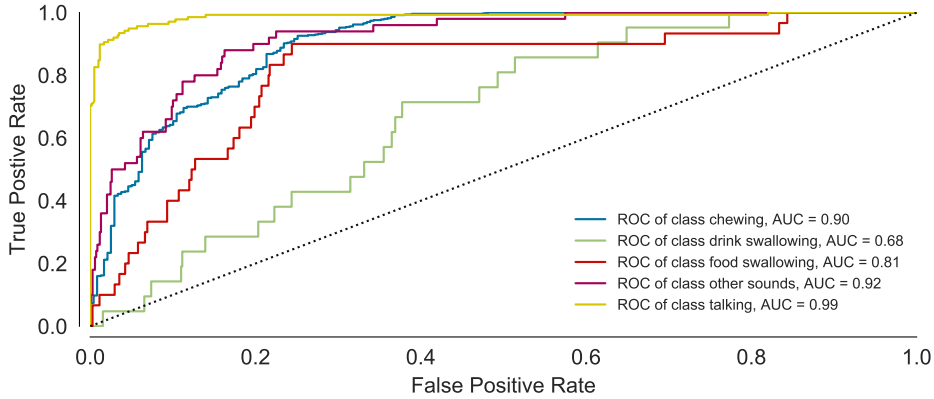
contrary, as shown in Fig. 5b, the model trained on the oversampled dataset performs much better: most of the positive instances are correctly predicted and very few negative instances are misclassified. Furthermore, this model has a high performance for all classes. The AUC represents the area under the entire ROC curve. A perfect classifier would correctly classify all positive instances and misclassifies no negative instances. Consequently, it would be a single point $P = (1, 0)$ and its $AUC = 1$. On the contrary, the worst classifier would incorrectly misclassify all positive and negative instances and would be represented with a single point $P = (1, 0)$ with an $AUC = 0$. The model trained on the oversampled dataset has much better performance because its $AUC > 0.98$ for all classes.

The models also achieved relatively high precision and high recall. The precision (Eq. (17)) evaluates how often a given sample that was predicted as positive was actually positive. The recall (Eq. (18)) estimates how many times a positive sample was predicted as positive.

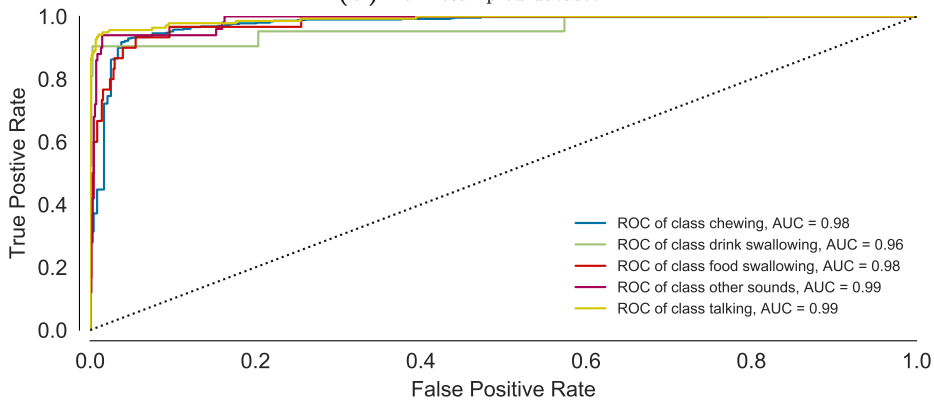
$$precision = \frac{TP}{TP + FP} \tag{17}$$

$$recall = \frac{TP}{TP + FN} \tag{18}$$

In short, the recall measures how many proper predictions the model makes while the precision estimates the pertinence of the predictions. For imbalanced datasets, the most optimum model would achieve a high recall without shrinking



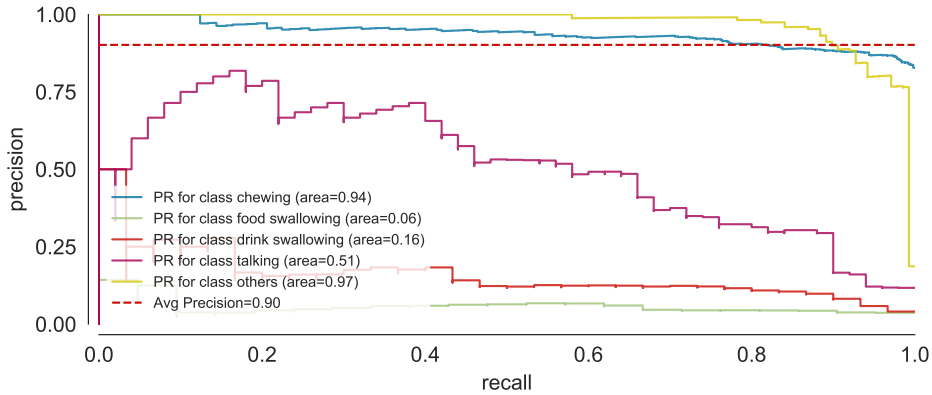
(a) non-resampled dataset



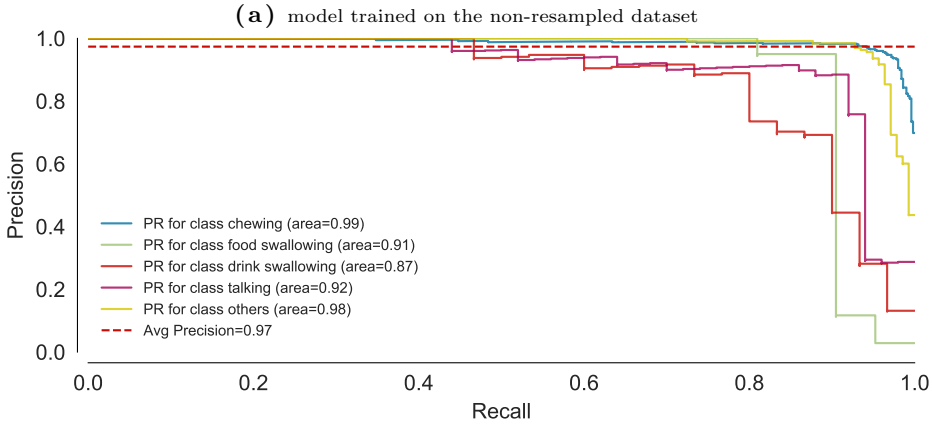
(b) resampled dataset

FIG. 5: Trade-off between the model's sensitivity and specificity the dotted line represents the performance of a classifier that generates predictions uniformly at random. The area under the curve (AUC) expresses the relationship between false positives and true positives. The model performed very well because it achieved a high AUC. It also has few false positives and predicted many true positives

its precision [27]. These two objectives, however, are often antagonistic because increasing TP of the minority classes also increases its number of FP, which reduces the precision. Fig. 6 shows the trade-offs between recall and precision. A perfect model ($area = 1.0$) would be made of two lines: a horizontal line $y = 1$ and a vertical line $x = 1$. Conversely, a poor classifier would only achieve a high recall only if its precision is low. Fig. 6 highlights the performance of the oversampled model compared to that of the non-resampled model. Indeed, the former (Fig. 6b) has a higher area and its curves are steepest when their recalls are closer to 1.0. On the contrary, the latter (Fig. 6a) has much lower areas, and for some classes (e.g.,food swallowing and drink swallowing) the performance is worse than a random classifier.



(a) model trained on the non-resampled dataset



(b) model trained on the resampled dataset

FIG. 6: Trade-offs between the recall and precision
The model trained on the non-resampled dataset has a much lower performance (e.g., low area) compared to model trained on the oversampled dataset. The latter has higher area and its curves are steepest when their respective recalls curves are closed to 1.0.

The F_β score (Eq. (19)) expresses the performance of a given classifier in terms of the compromise between the classifier's recall and its precision [27].

$$\begin{aligned} F_\beta &= (1+\beta^2)\cdot\frac{precision\cdot recall}{(\beta^2\cdot precision)+recall} \\ &= \frac{(1+\beta^2)\cdot TP}{(1+\beta^2)\cdot TP+\beta^2\cdot FN+FP} \end{aligned} \tag{19}$$

When $\beta = 0$, the F_β score would only consider the precision. A $\beta > 1$ gives more weight to recall over precision. In this study, $F_\beta(\beta = 0.5)$ and $F_\beta(\beta = 1)$ are considered. The former gives emphasizes precision over recall. The later gives an equal balance between the two. As shown in Table III, Table IV and Table V,

the model trained on the oversampled dataset outperformed the the one trained on the original dataset.

The Jaccard score (Eq. (20)) estimates the fraction of elements that are shared between two sets A and B. It is a ration between the number of times a given value occurs in both A and B to the total number of distinct items in A and B.

$$\begin{aligned} J(A,B) &= \frac{\left|A \cap B\right|}{\left|A \cup B\right|} \\ &= \frac{\left|A \cap B\right|}{|A| + |B| - \left|A \cap B\right|} \end{aligned} \tag{20}$$

where $\cap$ represent an intersection, $\cup$ denotes a union, and $||$ stands for the number of elements of the set. Consequently, in our case, it estimates the performance of the classifiers by comparing the predicted labels to the expected label. A perfect classifier would achieve a $Jaccard\,score = 1$. As shown in Table III, the predicted labels are 89% similar to the expected labels.

Unlike the other model evaluation measures, the Matthews Correlation Coefficient (MCC) (Eq. (21)) evaluates the performance of a classifier in terms of the four sections of the confusion matrix (i.e.,in terms of TP, FP, TN and FP).

$$\frac{c \times s - \sum_k^K p_k \times t_k}{\sqrt{\left(s^2 - \sum_k^K p_k^2\right) - \left(s^2 - \sum_k^K t_k^2\right)}} \tag{21}$$

with:
c the total number of sample correctly predicted
s the total number of samples
t_k the number of time a class k truly occurred
p_k the number of time class k was predicted

Compared to other model evaluation metrics, the MCC is more reliable because it produces a high score only when the model has good performances in all the four categories of the confusion matrix and is considered a good evaluation metrics even for imbalanced classes [28]. A mediocre classier has MCC values that range between -1 and 0 while a perfect classifier has a $MCC = +1$. Table III and Table V show that the models trained on the oversampled dataset have a good performance ($MCC = 0.85$). However, the model achieved the best performance when predicting the *talking* classes ($MCC = 0.93$) but blunders when predicting the *drink swallowing* classes ($MCC = 0.76$). On the contrary, the model trained on the non-resampled dataset has poor performance (Table IV), especially when predicting the *drink swallowing* classes ($MCC = 0.22$).

Finally, Fig. 7 shows the performance of the model with respect to the increase in the training samples. With much fewer samples, there is a large gap between the performance of the training and the test set. This may implies that the model has a high variance and is over-fitting. On the contrary, increasing the training sample closes the gap. However, with the available training samples, the training and the test score have not converged together, which implies that more training samples might lead to a better performance.

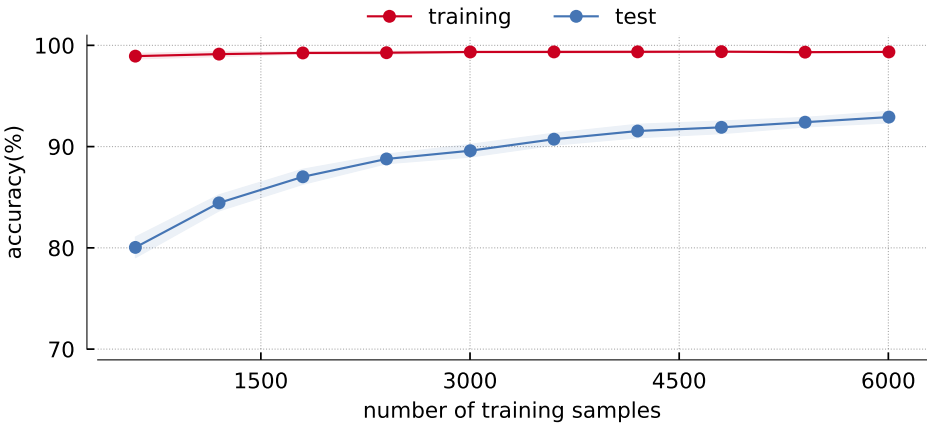


FIG. 7: Training and test score versus the number of training samples
The solid line indicates the mean value while the shaded area represents the standard deviation.Both the training and the test score exhibit a low variability; thus, the model has a a low bias and a low variance. However, the test score improves with more training samples, which implies that more training samples might lead to a better performance

4 CONCLUSION

Predominant methods for fighting against obesity focus on healthy eating but often ignore the importance of slowly eating and properly chewing in fighting obesity. This research proposed a method that recognizes common human actions during mealtime (i.e., food chewing, food swallowing, drink swallowing, and talking). The proposed approach uses a cheap wearable bone conduction microphone that records intra-body sounds that are produced when a person eats. We conducted experiments on 16 subjects. All experiments were conducted in free-living environments to capture the complexity of the real-world. For instance, some dietary sounds were recorded at home when the subjects were having supper with their families. Others were collected in a university cafeteria during lunch with colleagues. Additionally, there was no constraint on the type of food and the subjects were urged to have whatever they usually eat. We trained and tested machine learning models on the obtained dataset and found that the proposed approach achieves similar performances as that of the most recent state of the art models ($accuracy$ = 97.5%, $precision$ = 83.8%, $recall$ = 91.7%, $Jaccard\ score$ = 0.89, $Kappa\ score$ = 0.88, $F_1\ score$ = 87.2% and MCC = +0.85) even though our system uses modest machine learning models. Nevertheless, while the model generally performs well in distinguishing the various mealtime activities, it sometimes confused food and drink swallowing for chewing.

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